Experiment No.: 04

Title: Perform Clustering on an Appropriate Dataset

Batch: B3 **Roll No.: 16014223083** **Experiment No.: 04**

Aim: To perform clustering on a given dataset using the K-Means clustering algorithm and analyze the grouping of data points based on similarity.

Resources needed: VSCode, Python(NumPy, Pandas, Matplotlib, Scikit-learn), Google Colab, Jupyter Notebook

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Pre Requisite: Data Science, Machine Learning

Theory:

Clustering is an unsupervised learning technique used to group similar data points into clusters such that:

- Data points within the same cluster are highly similar
- Data points in different clusters are dissimilar

K-Means Clustering Algorithm

K-Means partitions the dataset into K clusters by minimizing the within-cluster sum of squares (WCSS).

Steps of K-Means Algorithm

1. Choose the number of clusters (K).
2. Initialize K centroids randomly.
3. Assign each data point to the nearest centroid.
4. Recalculate centroids as the mean of assigned points.
5. Repeat steps 3 and 4 until convergence.

Algorithm / Procedure

1. Load the dataset.
2. Select relevant features.
3. Normalize the data (if required).
4. Apply K-Means clustering.
5. Determine optimal K using Elbow Method.
6. Visualize clusters.
7. Analyze results.

Data Set
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

Iris Dataset

- Total instances: 150
- Attributes: Sepal length, Sepal width, Petal length, Petal width
- Type: Numerical dataset

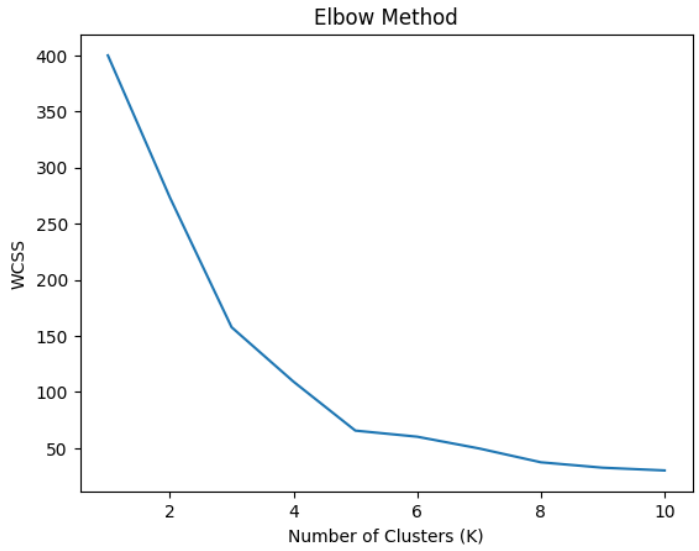
OUTPUT:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
# Select features for clustering
X = data[['Annual Income (k$)', 'Spending Score (1-100)']]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

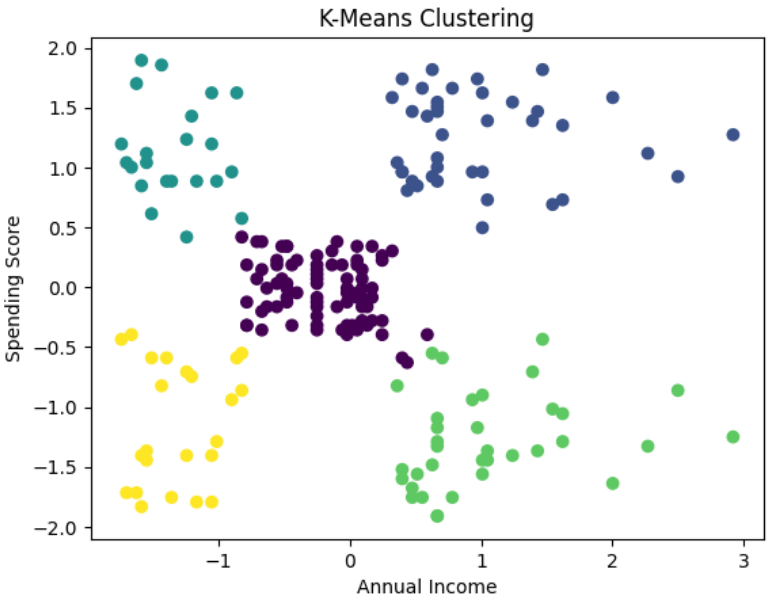
```
#Elbow Method (to find optimal K)
wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
```

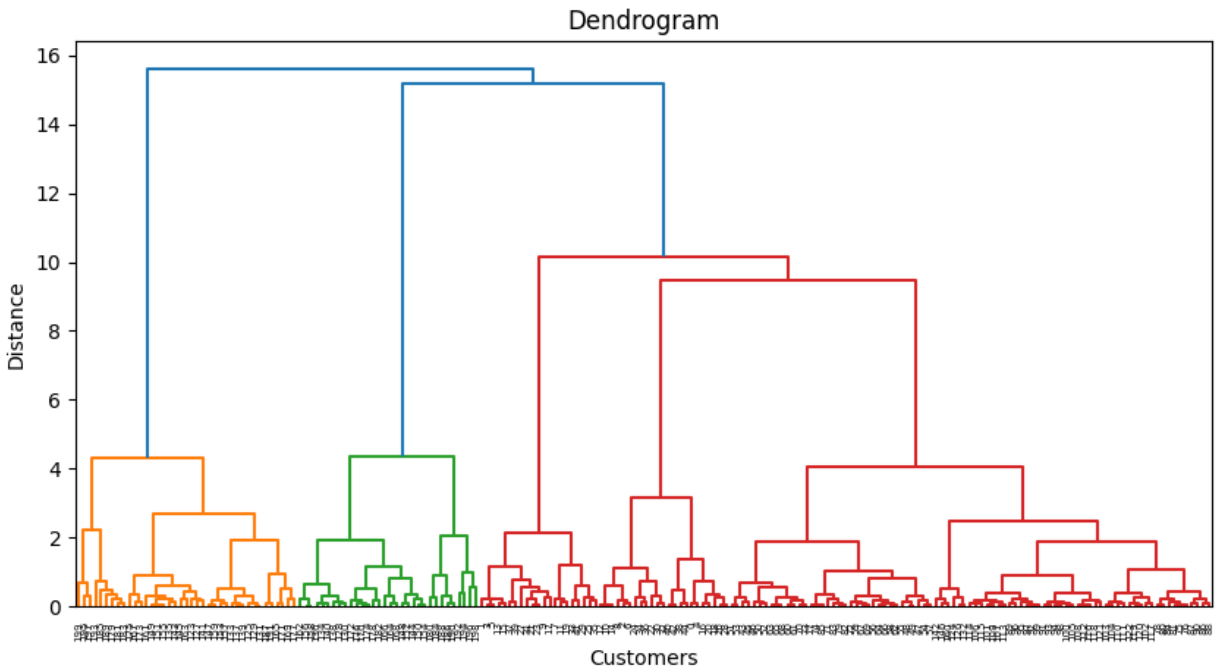
```
    wcss.append(kmeans.inertia_)
plt.plot(range(1, 11), wcss)
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```




```
#Apply K-Means (K = 5)
kmeans = KMeans(n_clusters=5, random_state=42)
kmeans_labels = kmeans.fit_predict(X_scaled)
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=kmeans_labels)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("K-Means Clustering")
plt.show()
```

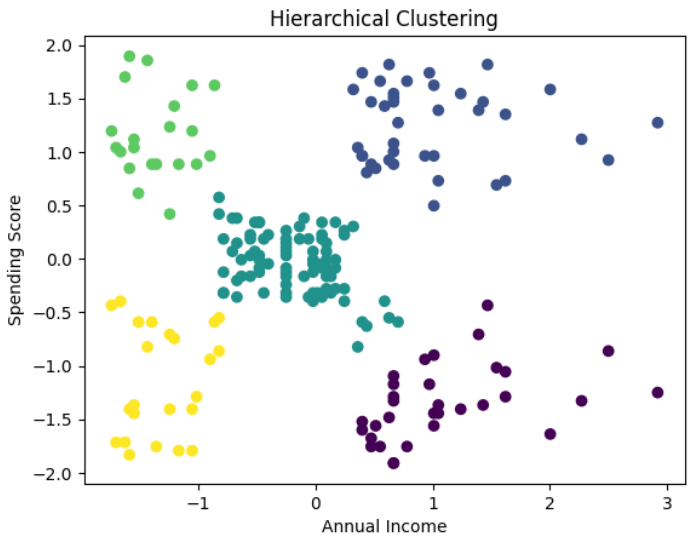



```
# Hierarchical Clustering
# Dendrogram
linked = linkage(X_scaled, method='ward')
plt.figure(figsize=(10, 5))
dendrogram(linked)
plt.title("Dendrogram")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()
```

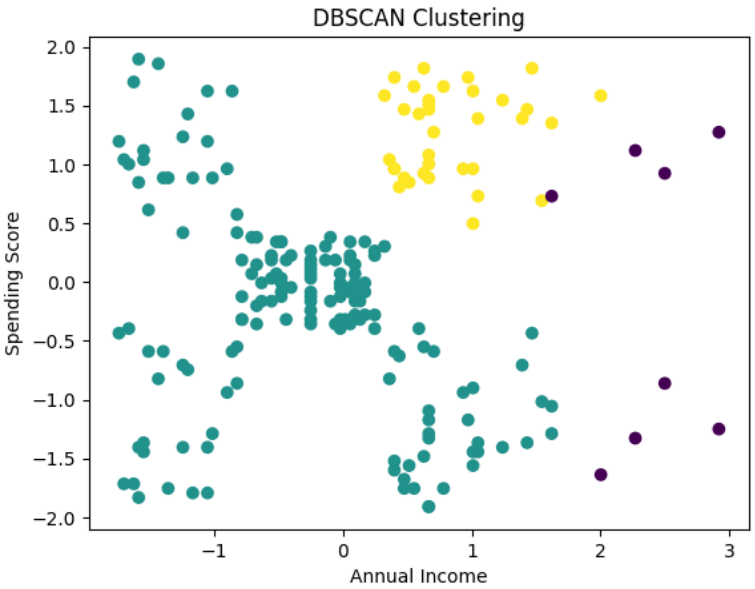


```
# Apply Agglomerative Clustering (K = 5)
hc = AgglomerativeClustering(n_clusters=5)
hc_labels = hc.fit_predict(X_scaled)

plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=hc_labels)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Hierarchical Clustering")
plt.show()
```




```
#DBSCAN
dbscan = DBSCAN(eps=0.5, min_samples=5)
dbscan_labels = dbscan.fit_predict(X_scaled)
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=dbscan_labels)
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("DBSCAN Clustering")
plt.show()
```



Outcomes:

CO3: Apply advanced analytics to predict and optimize decision making process

Conclusion: (Conclusion to be based on the objectives and outcomes achieved)

Clustering was performed on the Mall Customer dataset using K-Means, Hierarchical, and DBSCAN algorithms. K-Means and Hierarchical clustering formed clear customer segments based on income and spending behavior, while DBSCAN identified dense clusters and outliers. This experiment highlights the effectiveness of different clustering techniques for real-world customer segmentation.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

Websites:

1. *LearnVern: K-Means introductory example and theory*
2. *KDnuggets: Mall Customer Segmentation with K-Means Clustering example*
3. *Learnovita: K-Means algorithm steps and workin*